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Observing Alterations in Fault Dynamics Following Extended Production and Injection in a Carbonate Reservoir. A Case Study from UAE Offshore

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Abstract

Managing faults in carbonate reservoirs can be feasible with proper handling. Historically, various fault-related issues have been documented; some show no change, while others encounter significant problems like total mud losses, tight spots, stuck pipe, casing failure, washouts, breakouts, and hole collapse. This paper examines the characteristics of faults in their initial state and their behavioral changes after production/injection interference.

Seismically driven fault interpretation outlines the fault plane(s) location and dip/strike-slip components. To further reduce fault-associated risks, Allan diagrams help understand potential leakage through juxtaposition. This analysis reveals the initial characteristics of faults, aligned with seismic vintage data. At the well-planning stage, any intersected fault would be characterized using the same vintage data, indicating the depth and throw of the fault(s) but not recent behavioral changes. An example from the study area highlights the necessary predrill evaluation (while the fault was in its initial state) before drilling a producer well (Well-A). Well-A faced no difficulties during drilling nor any unexpected water breakthrough during production. An additional supportive data (e.g., borehole resistivity image) was obtained in Well-A to further define the fault/fracture zone. After several years of production with a consistent water cut, a new well (Well-B) was drilled through the same fault. While drilling near this fault zone, Well-B experienced complete mud losses. Upon Well-B intersecting the fault zone, fractures opened, causing the drilling fluid to choke Well-A, resulting in a complete cessation of its production. After a clean-out procedure, Well-A was revived; however, its water cut increased to over 50%.

Faults in the study area display more strike-slip than vertical components, resembling fractures or fracture corridors. The borehole resistivity image shows that most initial condition fractures/faults are closed, except for a few continuous conductive fractures. Although the predrill Allan diagram indicates some leakage through juxtaposition zones, Well-A's water cut remained minimal throughout its production phase. Prolonged production led to partial depletion of the fault zones, reducing the pore pressure gradient from 0.6 psi/ft to 0.5 psi/ft and the frac gradient from 0.8 psi/ft to 0.7 psi/ft. This reduction was significant within the fracture zones. When Well-A was reactivated, the water cut was initially low. This was anticipated because the opened fracture might close. After a few weeks of production, the water cut drastically increased from

10% to 62%. This significant increase in water cut suggests that a large amount of water was being produced along with the oil, likely due to the communication between the reservoirs and the opened fractures.

Accurately characterizing faults and fractures during well-planning is essential for optimal lower completion planning, which helps to prevent early water breakthrough and supports a sustainable development plan.

Introduction

Hydrocarbon reserves in the Middle East primarily come from carbonate platforms dating from the Upper Permian to Oligocene, with producing fields typically being low dip (~1-2 degrees) anticlines shaped by tectonic episodes in Late Cretaceous and Late Oligocene. The study covers a large offshore oil field in the UAE, focusing on Well-A's production and injection activities in nearby wells like Well-C in Reservoir B.

Water injections enhance hydrocarbon recovery and manage produced water disposal. Carbonate rocks with karst collapse features show high injectivity and minimal risks for reservoir damage. Faults in the area display strike-slip characteristics, resembling fractures, with borehole resistivity images showing closed initial condition fractures.

Fault analysis using seismic data and Allan diagrams revealed minimal leakage during Well-A's production. Proper fault management in carbonate reservoirs is crucial to avoiding issues like mud losses, wellbore instability, and water breakthrough. The pore pressure gradient originally at 0.6 psi/ft dropped to 0.5 psi/ft after years of production, and the frac gradient decreased from 0.8 psi/ft to 0.7 psi/ft, indicating increased susceptibility to opening under lower pressures.

The Hubbert & Willis equation and nomograph method are used to estimate frac gradients, factoring in overburden gradient, pore pressure gradient, and Poisson's ratio, noting that frac gradients generally increase with depth until abnormal pressure sections are reached.

Tectonic Episodes

The tectonics of Abu Dhabi can be characterized by three principal tectonic cycles:

Key tectonic episodes are indicated on the stratigraphic column in [Fig.1](#). The present-day subsurface structures primarily result from two significant tectonic events: one in the Late Cretaceous (characterized by WNW-ESE compression) and another in the Late Oligocene (characterized by NE-SW compression). Major faults, forming en-echelon zones, emerged due to the WNW-ESE compression during the Late Cretaceous period. These faults generally trend in a NW-SE direction and extend for several kilometers. During the Late Oligocene, tectonic stresses shifted to NE-SW, further deforming the subsurface structures. The cumulative impact of these tectonic episodes creates complexity in understanding the subsurface structures ([Ahmad et al., 2022](#)).

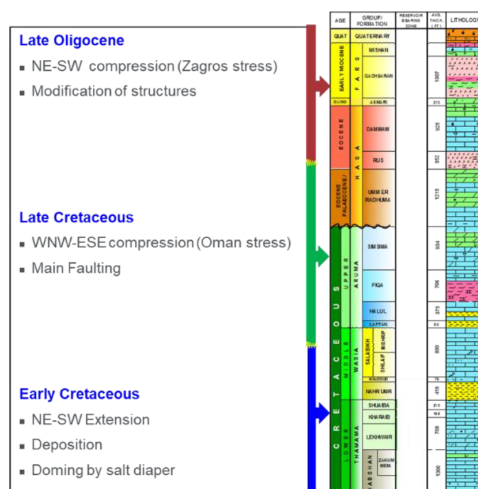


Figure 1—Generalized lithostratigraphic column showing rock units comprising reservoir and overburden within study area. Text on the left label's important tectonic episodes (Ahmad et al., 2022)

Key Structural Elements

The structural elements of Abu Dhabi have been influenced by various tectonic events over geologic time (Noufal et al., 2016), including:

- North-South Basement Lineaments: These lineaments, known as Arabian trends, were formed during the Pre-Cambrian era.
- Mender/Lekhwair High and Qatar Arch: Structural growth in these regions has played a significant role in the area's geologic history.
- Extensional Phase and Basin Subsidence: This phase marked the subsidence of the area, forming various basins.
- Compressional Period: During the Campanian-Maastrichtian period, the Oman Orogeny caused compressional forces, leading to the emplacement of nappes.
- Oman Mountains Uplift: The uplift of these mountains in the late Tertiary period significantly altered the region's geology.
- Zagros Orogeny: Late Tertiary-Quaternary events, including the opening of the Red Sea and the north/northeast movement of the Arabian plate against the Turkish and Iranian plates, had an impact on the area's tectonics. This movement resulted in a subduction zone.
- Halokinesis: The movement of the Hormuz infra-Cambrian salt basin has been continuous since the Triassic period, particularly affecting the offshore areas.
- Rub al Khali Basin Subsidence: This geological event has been occurring since the Triassic period, alongside the subsidence of the Ras Al Khaimah basin during the late Tertiary.

Seismic Interpretation and Fault Characterization

Surface seismic is a primary dataset for evaluating subsurface areas away from boreholes; however, the integration of seismic data with actual well data remains crucial. Seismic noise continues to be a significant impediment in this context.

In the study area, a low fold (~300) 3D Ocean Bottom Cable (OBC) seismic survey (using hydrophone and geophone) was conducted in shallow water depths ranging from approximately 5 meters to 30 meters. Two main challenges were identified during seismic data acquisition: the first challenge pertains to near-surface distortions, which generate various types of seismic acquisition noises, including ambient noise,

reverberations, and diffractions. The second challenge involves the impact of overburden geological noise, which can descend to the reservoir level and cause substantial pseudo-structural deformation.

Near-surface impact is primarily attributed to variations in water depth, receiver coupling, and seabed reflectivity, ultimately reducing coherent reflectivity on gathers and consequently on the final stack. Additionally, misunderstandings regarding overburden geology can result in significant misleading impressions at the target reservoir level (Ahmad et al., 2022).

Fault interpretation driven by seismic data involved identifying fault planes and analyzing their dip and strike-slip components. Detailed seismic analysis enabled the identification of key fault characteristics, such as depth, displacement, and orientation. The study revealed that faults in the area predominantly exhibit strike-slip movements rather than vertical displacements, resembling fractures or fracture corridors. This insight was crucial for planning drilling operations and managing reservoir performance. Well-A intersects the fault zone near its total depth (TD). On surface seismic data, faults appear as minor undulations due to minimal displacement. Faults are traced across lines, forming an en-echelon pattern visible on the map. The en-echelon reference pattern is also illustrated to demonstrate small-scale variations.

A Spiral 3D VSP survey was conducted using 70 level 3C receivers placed at 20 m intervals in a deviated well, with the top 20 receivers positioned within tubing. The shot point interval was set to 25 m, the radial distance between spiral arms was 50 m, the maximum source offset reached 2,916 m, and a total of 18,364 shots were carried out using an airgun source boat. Additionally, a Zero-Offset VSP was recorded up to the surface to provide essential input for 3D VSP imaging velocity model building. The overall quality of the 3D VSP data was assessed to be good. Near offset source frequencies recorded up to 120 Hz and far offset frequencies up to 70 Hz, which is higher than those observed in surface seismic data. To obtain a P-wave image, the Generalized Radon Transform (GRT) migration approach (Miller et al., 1987), theoretically similar to Kirchhoff migration, was applied (Fig. 2). This method resulted in comprehensive subsurface image coverage that will assist in future time-lapse 3D VSP monitoring. Fault imaging showed slight improvements on the 3D VSP compared to surface seismic data.

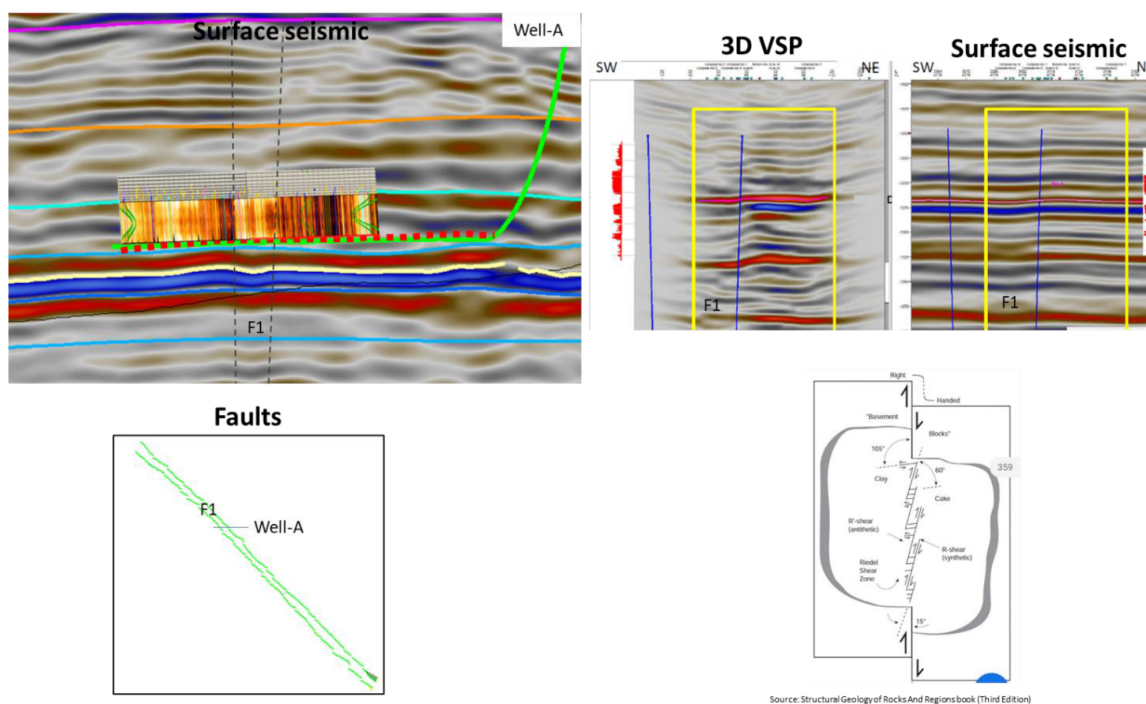


Figure 2—Well-A intersects the fault zone near the total depth (TD). On surface seismic data, faults are indicated by small undulations due to minimal fault's throw. 3D Vertical Seismic Profiling (VSP) demonstrates a slight improvement in fault delineation. The borehole resistivity image clearly demarcates the fault.

Advanced borehole resistivity imaging technique was employed to achieve real-time monitoring of fault and fracture conditions. This method facilitated accurate tracking of fault activity and variations in fracture conductivity, essential for informed decision-making during drilling and injection operations. Borehole resistivity imaging revealed that most fractures and faults were initially closed, with only a few showing continuous conductivity (Fig. 2). The integration of seismic data with borehole imaging played a crucial role in developing adaptive management strategies aimed at mitigating risks associated with evolving fault behavior, such as early water breakthrough and mud losses.

Allan diagrams were extensively utilized to analyze leakage potential through juxtaposition zones, providing a detailed visualization of fault behavior and its interaction with surrounding rock formations. This initial analysis yielded valuable insights into the fault characteristics, identifying areas with higher leakage potential and those with more stable conditions based on seismic vintage data. The Allan diagram assessments conducted during the production phase of Well-A indicated that the leakage potential was minimal, primarily due to the closed nature of most fractures and the minor fault throw (Fig. 3).

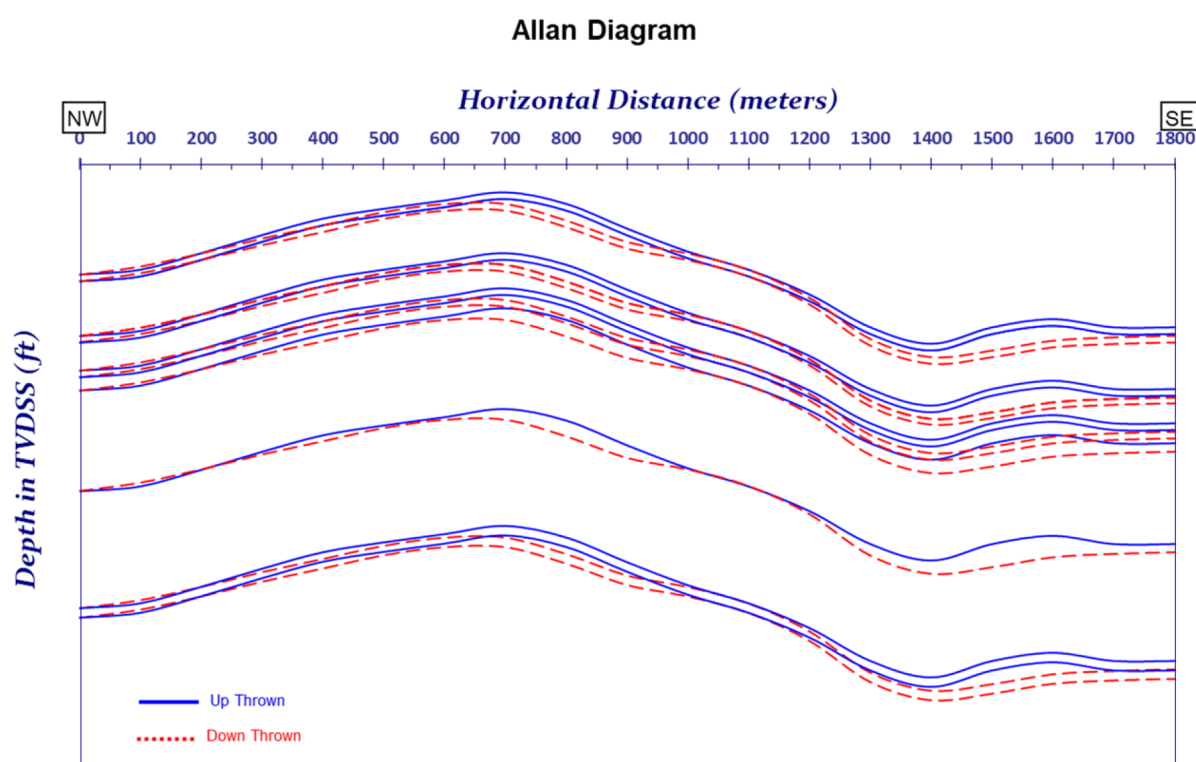


Figure 3—The Allan diagram of Fault F1 indicates minimal fault throw, suggesting a low potential for leakage against juxtaposition rocks.

Well performance and Drilling

Well-A was drilled and completed in Reservoir A without encountering significant drilling issues, successfully reaching the target reservoir intervals. This well intersected Fault F1 near its total depth (TD) and commenced production with a stable oil rate and a 10% water cut (Fig. 4). Well-C, an injector well, was drilled into the deeper Reservoir B and has demonstrated a consistent injection performance, with an average rate of 12,000 STB/D (Fig. 5).

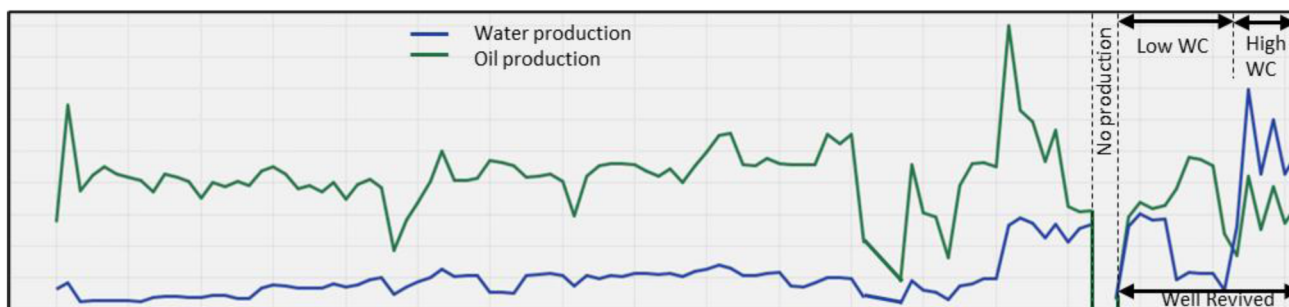


Figure 4—The performance chart for Well-A illustrates consistent production with a low water cut (10%). Upon Well-B intersecting the fault zone, fractures opened, causing the drilling fluid to choke Well-A, resulting in a complete cessation of its production. After a clean-out procedure, Well-A was revived; however, its water cut increased to over 50%.

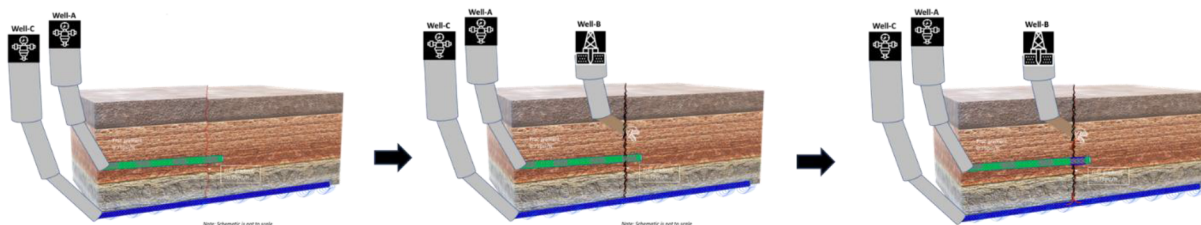


Figure 5—Schematic of potential cross-reservoir water movement from well-C to well-A in the vicinity of Fault (F1) zone following drilling of well-B with mud weight, exceeding the fracture gradient of reservoir a

Five years later, Well-B was planned to be drilled and completed in Reservoir-B, crossing the same fault F1 encountered by Well-A. During the drilling of Well-B, significant mud losses occurred due to overbalance pressure, which led to the opening of fractures and faults. As a result of these losses in the fractured zone while Well-A was in production, all the drilling fluid entered Well-A, causing it to stop producing completely. However, after a clean-out procedure, Well-A resumed production in a few days. Due to the opening of the fracture/fault zone, it is now anticipated that Reservoir-A and Reservoir-B are interconnected. When the well was activated at previously maintained rates, the initial water cut was low, possibly because the opened fracture might close. After a few weeks of production, the water cut significantly increased from 10% to 62% (Fig. 4).

Pore pressure and Fracture gradient

Following prolonged production, the fault zones experienced partial depletion, resulting in a reduction in pore pressure and consequently the frac gradient. The initial pore pressure gradient was 0.6 psi/ft, which dropped to 0.5 psi/ft after years of production. Consequently, the frac gradient decreased from 0.8 psi/ft to 0.7 psi/ft. This reduction was particularly notable within the fracture zones. The drilling of Well-B using standard mud weight increased the overbalance pressure, causing fractures and faults to open, leading to complete mud losses during drilling operations in Reservoir-A, especially when intersecting Fault F1. Furthermore, this change led to a significant increase in water cut for Well-A, observed during the drilling operations of Well-B, which was interconnected with Well-C. These observations underscore the sensitivity of fault zones to pressure variations and highlight the need for adaptive management strategies to maintain wellbore stability and reservoir integrity.

Fracture gradient can be estimated using two methods. The first method is based on the equation proposed by Hubbert and Willis: $F = 0.5(1 + P/D)$, where F represents the fracture gradient in psi/ft and P/D denotes the pore pressure gradient in psi/ft. Although this equation tends to predict values that are lower than those observed in field data, it remains valid as it indicates that the fracture gradient increases only when the pore pressure gradient increases. However, empirical evidence suggests that the fracture gradient generally

increases with depth, irrespective of pressure behavior, until the abnormal pressure section is encountered where the fracture gradient may decrease (Ben et al., 1969).

Another method involves estimating the fracture gradient using a nomograph (Fig. 6), which requires inputs such as the overburden gradient, pore pressure gradient, and Poisson's ratio.

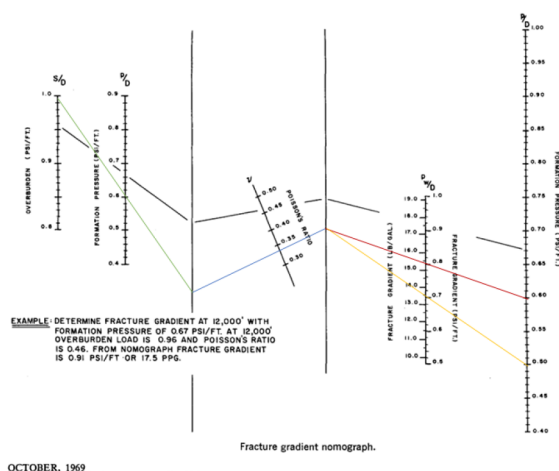


Figure 6—Fracture gradient nomograph (Ben A. Eaton) shows an Initial formation pressure was 0.6psi/ft and initial fracture gradient of 0.82 psi/ft for matrix limestone (based on a Poisson ratio of 0.33 and overburden gradient of 1.0 psi/ft). After years of production, formation pressure drops to 0.5 psi/ft, lowering the fracture gradient to 0.7 psi/ft. The gradient is expected to drop faster in fault zones.

Conclusion

Well-A was drilled and completed in Reservoir-A without significant issues, intersecting Fault F1 near its total depth (TD). It began production with a stable oil rate and a 10% water cut. Well-C, an injector well, was drilled into the deeper Reservoir-B and had a good injection performance with an average rate of 12,000 STB/D. Five years later, Well-B was drilled and completed in Reservoir-B, crossing the same Fault F1 encountered by Well-A. During the drilling of Well-B, significant mud losses occurred due to overbalance pressure, leading to the opening of fractures and faults. This caused all the drilling fluid to enter Well-A, choking it and ceasing its production. After a clean-out procedure, Well-A was revived. The opening of the fracture/fault zone indicated communication between Reservoir-A and Reservoir-B. Initially, the water cut was low, but it drastically increased from 10% to 62% after a few weeks of production.

Prolonged production led to partial depletion of the fault zones, reducing the pore pressure gradient from 0.6 psi/ft to 0.5 psi/ft and the frac gradient from 0.8 psi/ft to 0.7 psi/ft. This reduction was significant within the fracture zones. The drilling of Well-B with standard mud weight increased the overbalance pressure, causing fractures and faults to open, resulting in complete mud losses in Reservoir-A, especially when intersecting Fault F1. This also led to a substantial increase in water cut for Well-A during the drilling operations of Well-B, which was interconnected with Well-C. These changes highlighted the vulnerability of fault zones to pressure variations, emphasizing the need for adaptive management strategies to maintain wellbore stability and reservoir integrity.

The frac gradient is estimated using two equations: Hubbert & Willis [$F=0.5(1+P/D)$] and a nomograph requiring inputs such as overburden gradient, pore pressure gradient, and Poisson's ratio. The Hubbert & Willis equation predicts values that are usually too low compared to field data, but it is not invalid. Experience shows that the frac gradient increases with depth, regardless of pressure behavior, until the abnormal pressure section is traversed, where the frac gradient may decrease.

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Nomenclature

FG =Fracture Gradient

P/D = Pore Pressure Gradient

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